

Problem Statement Selection

Problem statement addressed: Cleveland Clinic — *Unlocking undruggable targets: quantum simulation of allosteric signal propagation* (Global Quantum + AI Challenge 2026, Cleveland-Clinic-Challenge-Statement-vF-1.pdf).

How we interpreted it

Rather than only running the requested algorithm on the provided structures, our Phase-1 work audited the challenge’s own premise and dataset. Our measurements show that the standard benchmark targets conflate **static cavity retrieval** with **genuine cryptic-pocket discovery**: roughly half of apo/holo pairs cannot express the required contrast at all.

Our proposal therefore addresses the challenge’s core objective by delivering a **validated, unfounded benchmarking instrument** capable of certifying allosteric methods — quantum or classical — alongside the required predictions.

Targets

We address the complete mandatory set required by Challenge Statement §6: **KRAS G12C, BCR-ABL1, cardiac myosin, and c-Myc**.

Three structures deviate from Table 1 (4LDJ replacing 4OBE; 8QYP/8QYR replacing 5TBY/6C1H); BCR-ABL1’s 10PL is retained, not substituted. **Each is stated, with its reason and its evidence, in Section 1 of the Concept Proposal**, and each traces to the organisers’ clarification of 2026-08-26. They are not repeated here: a single authoritative statement is less likely to drift than two, and the Concept Proposal is the document a reviewer assesses.

Required outputs

Challenge Statement §5’s three deliverables are supplied as follows:

§ 5 output	Where
Connectivity Matrix — N×N quantum connectivity strength	Connectivity_Matrices.csv
Hit List — top five predicted allosteric residues per target	Solution_Outputs.pdf , and machine-readable in the same file set
Methodological Report — the quantum metric chosen, and why it proxies biological signal transmission	Section 2 of the Concept Proposal , with per-target detail in Solution_Outputs.pdf